Non-Axiomatic Reasoning for Longitudinal Identity Resolution: NARS on ICE-ID and Standard ER Benchmarks

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Abstract

We evaluate OpenNARS-for-Applications (ONA) for longitudinal identity resolution on Icelandic historical census data spanning 220 years (1703–1920), while benchmarking it on standard entity resolution datasets (Abt–Buy, Amazon–Google, DBLP–ACM, DBLP–Scholar). NARS is a general-purpose AI framework designed to reason with limited knowledge and resources; its core is Non-Axiomatic Logic (NAL), a term-based logic. Our experiments show that this evidence-based symbolic matcher provides a transparent and auditable ER baseline while remaining competitive on ICE-ID and bibliographic ER benchmarks. In the deployed LLR scoring variant, each prediction is computed as a decomposable sum of named symbolic judgments with explicit evidence weights, so explanations follow directly from the scoring function. We report pairwise metrics, calibration sensitivity, and a deployment-faithful candidate-graph evaluation on ICE-ID, and compare against classical probabilistic and neural baselines across standard ER datasets.

1 Introduction

Linking historical census records is a core operation in research on social mobility, demographic change, migration, and epidemiology, but it remains difficult in practice because names vary across time, fields are missing, and administrative boundaries change [26, 1]. Supervised record-linkage systems have improved performance in specific settings, including nineteenth-century U.S. censuses [4], though the choice of training data and evaluation protocol significantly affects reported accuracy [5]. Yet most benchmarked setups still simplify the problem by using short temporal windows, limited relational structure, and flat geographic representations [23].

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This paper asks whether a symbolic reasoning system can remain competitive in longitudinal identity resolution while improving transparency at the level of individual match decisions and error analysis. We evaluate a Non-Axiomatic Reasoning System (NARS), implemented with OpenNARS-for-Applications (ONA), on ICE-ID and compare it against standard entity resolution (ER) benchmarks. ICE-ID provides a long-horizon test setting: it spans 16 Icelandic census waves from 1703 to 1920, contains 984,028 records, and includes 226,864 expert-curated person clusters. It introduces temporal drift, hierarchical geography, and recurring patronymic name collisions, which makes it a useful benchmark for both symbolic and non-symbolic matchers. ICE-ID also complements existing historical and longitudinal linkage infrastructures such as the IPUMS linked samples [16, 17] and the North Atlantic Population Project [18, 28].

We study two tasks on ICE-ID:

1. **Intra-census linkage**: determine whether two records from the same census refer to the same person.
2. **Cross-census linkage**: determine whether two records from different censuses refer to the same person despite spelling variation, age progression, and movement across locations.

All ICE-ID experiments use a strictly temporal split: records up to 1870 for training, 1871–1890 for validation, and 1891–1920 for testing. We report pairwise metrics (precision, recall, F_1, ROC-AUC) and clustering metrics (Adjusted Rand Index, ARI) to separate local pair decisions from global entity consistency.

We compare three families of approaches under this benchmarking setup:

- *Classical probabilistic linkage*: Fellegi–Sunter and deterministic rule-based matchers;
- *Symbolic reasoning*: NARS, which scores record pairs through accumulated evidence over named symbolic judgments, yielding a score that can be decomposed into auditable positive and negative contributions;
- *Neural and LLM-based matchers*: Ditto [13], HierGAT [33], zero-shot methods such as AnyMatch [34], and LLM-based matching methods (including the MatchGPT-style baseline used in our experiments), which can be strong predictors but do not expose the deployed score in the same symbolic evidence vocabulary.

Our results show that NARS provides a competitive and intrinsically interpretable baseline. On ICE-ID, it reaches pairwise performance comparable to strong probabilistic linkage baselines and close to the best deep learning result reported in our benchmark suite, while preserving a decomposable decision trace for each pair through explicit judgment contributions. On standard ER datasets, NARS performs especially well on bibliographic matching (DBLP–ACM, DBLP–Scholar), while remaining weaker on product-matching datasets (Abt–Buy, Amazon–Google), where long noisy text favors learned representations. These results support the use of symbolic reasoning as a practical ER baseline when auditability is a first-class requirement and motivate hybrid pipelines that combine symbolic evidence with neural representations.

2 Methods

2.1 ICE-ID Benchmark

We evaluate on ICE-ID [9], a benchmark of 984,028 Icelandic census records (1703–1920) with 226,864 expert-labeled person clusters. We use the canonical temporal splits (up to 1870 train / 1871–1890 validation / 1891–1920 test) and report pairwise metrics (Precision, Recall, F_1, AUC), clustering metrics (ARI, B^3), and ranking metrics (P@k, R@k). Full dataset documentation follows the datasheet and data-statement reporting norms of Gebru et al. [7] and Bender and Friedman [2] and is provided in the companion data paper.

2.1.1 Data Preparation

Records are loaded with string columns (names, status labels) normalized to lowercase and numeric fields coerced to integers. The key numeric columns are `id` (unique record identifier), `heimild`

(census wave, i.e. the year of enumeration), `birthyear` (year of birth), `person` (expert-curated identity label linking records of the same individual), and geographic/kinship foreign keys (`farm`, `parish`, `district`, `county`, `partner`, `father`, `mother`). Missing IDs are dropped; missing name fields are filled with empty strings. NARS and the symbolic matchers operate directly on these record pairs; the machine learning (ML) ensemble baseline uses pairwise TF-IDF features over concatenated record text. All ICE-ID experiments under the shared protocol use the same raw record representation and temporal splits keyed by `heimild` (census wave).

2.1.2 Pair Construction and Sampling

Positive pairs are drawn from expert-labeled clusters: two records are labeled as a match if they share the same `person` ID. For *cross-wave* linkage, pair generation spans distinct census-year windows while preserving the same cluster semantics. Negative pairs are sampled from records assigned to different clusters, using a 2:1 negative-to-positive ratio within each split.

For ICE-ID, pair generation and any subsampling are performed *inside* the fixed temporal splits (train: up to 1870, validation: 1871–1890, test: 1891–1920); no random split is applied across years. If a split contains more than 50,000 candidate pairs, we uniformly subsample to that cap so that NARS and the ML baselines are evaluated on the same bounded pair set. The cap is configurable, and the implementation does not impose a same-census-year constraint on negatives; within-wave evaluation can be obtained by restricting the record pool to a single `heimild` in a given run.

2.2 ML Ensemble Baseline

As a baseline, we compare against an ensemble of four tree-based classifiers (XGBoost, LightGBM, CatBoost, Random Forest) averaging match probabilities. The ensemble is trained on pairwise text-derived and structured comparison features computed from the same raw record pairs used by NARS. Model selection uses 10-fold cross-validation within the training split; final evaluation uses the same held-out validation and test splits as NARS.

2.3 OpenNARS-for-Applications

Implementation: We use OpenNARS-for-Applications (ONA) [8] as the underlying NARS engine, building on the Non-Axiomatic Logic (NAL) framework [30]. ONA defines the judgment vocabulary and evidence framework; record pairs are converted into Narsese statements encoding attribute agreements and disagreements. The reported experiments use the LLR scoring instantiation described in the "Practical scoring variant" paragraph below.

Algorithm Overview: Figure 1 summarizes the procedure.

```
Inputs:  labeled training pairs (r1, r2, y), pool size P, patterns used n

1) For each training pair:
  J = preprocess(r1, r2)                                    (atomic judgments)
  tv = TruthValue(f=y, c=0.9)
  add/update (J → tv) in PatternPool (evict if > P)

2) Calibrate a threshold τ on held-out labeled pairs
  (ICE-ID: median midpoint; classic ER: validation F1)

3) For each test pair:
  Jq = preprocess(r1, r2)
  select n reference patterns from pool (top/bottom expectation)
  score = revise evidence from overlaps between Jq and selected patterns
  predict match if score ≥ τ
```

Figure 1: Conceptual NARS entity resolution procedure (pseudocode). Reported experiments use the LLR scoring variant described below.

This pseudocode illustrates the conceptual pattern-pool formulation for intuition; the reported implementation replaces Step 3's revision-based scoring with the LLR scorer described below.

Complexity: Conceptual pattern-pool scoring is $O(N \cdot n \cdot |J|)$ where N = pairs, n = reference patterns, $|J|$ = judgments per pair; the LLR variant reduces to $O(N \cdot |J|)$ since each judgment maps to a precomputed log-ratio.

The LLR scoring pipeline used in all reported experiments proceeds as follows:

1. **Judgment Generation.** A preprocessing function converts each record pair (r_1, r_2) into a set of atomic judgments—e.g., `same_nafn_norm` (normalized full-name match; *nafn* is Icelandic for "name") and `birthyear_close` (birth years within 5 years)—based on attribute comparisons for name, birth year, sex, census wave (`heimild`), and location.
2. **Evidence Accumulation.** During training, each labeled pair increments per-judgment positive or negative counts (with Laplace $+1$ smoothing). The accumulated counts yield a log-likelihood ratio $\mathrm{LLR}_j = \log \frac{P(j|\text{match})}{P(j|\text{non-match})}$ for every judgment type j.
3. **Pair Scoring.** For a new pair, the generated judgments are looked up in the LLR table, summed with a base-rate log-prior, and passed through a sigmoid to produce a match probability.

Interpretability of the deployed scorer. In the LLR pipeline, the pair score is an additive function of named judgment tokens, so each prediction can be decomposed into exact per-judgment contributions to the final log-odds. This makes local explanations (why a pair matched), global analysis (which judgments dominate), and counterfactual inspection (how the score changes when a judgment is removed) native to the deployed model rather than a separate post-hoc analysis layer.

2.3.1 Conceptual NARS Formulation (Background)

The following pattern, pattern-pool, and revision rules describe the full conceptual NARS formulation that motivates the design and establish the correspondence with Non-Axiomatic Logic (see Appendix A.4). The reported experiments use the LLR scoring pipeline above, which is a practical instantiation of this NARS framework that replaces the explicit pattern-pool revision step with precomputed per-judgment log-likelihood ratios.

Pattern. A pair of census records yields an atomic pattern: the unordered set of all generated judgments, annotated with truth-value $(1, 0.9)$ if the records belong to the same individual or $(0, 0.9)$ otherwise.

Pattern Pool. The pattern pool contains patterns, sorted in ascending order of the expectation of the truth-value.

Inference Rule. Given two patterns p_1 and p_2 containing judgment sets j_1 and j_2 with truth-values t_1 and t_2, the proposed inference rule derives three new patterns:

1. The judgment set of the new pattern is $j_1 \setminus j_2$, and the truth-value is t_1.
2. The judgment set of the new pattern is $j_2 \setminus j_1$, and the truth-value is t_2.
3. The judgment set of the new pattern is $j_1 \cap j_2$, and the truth-value is $\mathrm{revise}(t_1, t_2)$.

These rules ensure that each derived truth-value is inherited from parent evidence rather than fabricated. In case 1, evidence contributing to t_1 also supports the new sub-pattern; its truth-value may later be revised with independently obtained patterns. Case 2 is symmetric.

In case 3, the intersection pattern is a sub-pattern of both parents, so both t_1 and t_2 contribute evidence and revision merges them. To avoid double-counting, only patterns derived from independent evidence sources may participate in a single revision step.

Learning as Recognition. In the conceptual formulation, the system constructs a query pattern and measures its overlap with existing patterns in the pool. The degree of overlap determines the match score. During this process, new patterns may be added to the pool and existing truth-values may be updated.

In the conceptual formulation, the final expectation score can be thresholded to obtain a binary match decision. In the reported implementation, threshold calibration is performed in the deployed LLR pipeline described below.

Conceptual truth-value scoring (not used in reported experiments). At scoring time, up to n reference patterns are retrieved from the pool—half with the highest expectation (likely matches) and half with the lowest (likely non-matches)—to provide balanced evidence. For each reference pattern, a pairwise truth-value is computed: frequency f equals the size of the judgment intersection divided by the size of the longer pattern, and confidence c is inherited from the longer (more specific) pattern. These pairwise truth-values are then merged via revision, and the final expectation e serves as the match score: higher e indicates stronger evidence that the two records belong to the same individual.

Practical scoring variant (summary). The LLR pipeline described above replaces truth-value revision with per-judgment log-likelihood ratios (Laplace-smoothed). The pair score is $\sigma\big(\sum_j \mathrm{LLR}_j + \log \frac{p}{1-p}\big)$, where σ is the logistic sigmoid and p is the training-set base rate. Table 2 documents the parameters.

2.3.2 Threshold Calibration (Deployed LLR Pipeline)

To convert LLR-based match probabilities into binary decisions, we calibrate a threshold on labeled validation pairs. For ICE-ID, after accumulating LLR statistics from training pairs, we score the held-out validation set to obtain separate positive and negative score distributions. We then set

$$\tau \;=\; \frac{\mathrm{median}(\text{pos_scores}) + \mathrm{median}(\text{neg_scores})}{2}.$$

At test time, any pair with score $\geq \tau$ is predicted as a match, otherwise as a non-match. For classic ER benchmarks, we instead select τ to maximize validation F_1 to handle extreme class imbalance.

Calibration Split: For ICE-ID, we compute τ on the held-out validation pairs, ensuring no leakage from test data.

Calibration Sensitivity: Table 1 reports four representative threshold strategies:

Table 1: Threshold calibration sensitivity on ICE-ID. All strategies achieve near-identical performance due to well-separated score distributions.

Strategy	$\mathbf{F_1}$	**Threshold**
Fixed 0.5	0.995	0.500
Train threshold	0.996	0.100
Platt scaling	0.995	0.500
Isotonic regression	**0.996**	0.500

On ICE-ID with well-separated positive/negative score distributions, all shown strategies achieve near-identical F_1 (≈0.995–0.996).

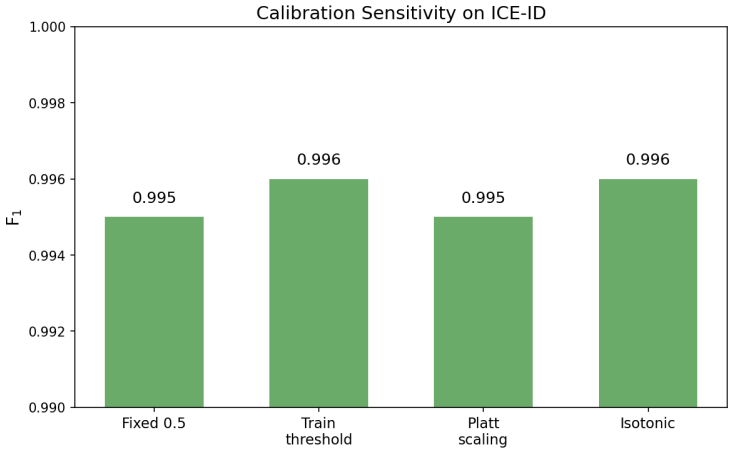

Figure 2: Calibration strategy comparison. All strategies achieve $F_1 > 0.99$ on ICE-ID due to well-separated score distributions.

Figure 2 mirrors Table 1: the spread between best and worst shown F_1 is about 0.001, so thresholding choice has limited impact under this ICE-ID setup.

2.3.3 Computational Resources

Experiments were run on an Intel® Core™ Ultra 9 185H system with 32GB RAM. The NARS benchmark runs reported here (10 runs across pairwise-benchmark and candidate-graph evaluation modes) took approximately 2 hours.

2.3.4 Hyperparameters and Reproducibility

Following the model-card reporting recommendations of Mitchell et al. [19], Table 2 documents the LLR scoring configuration used in all reported experiments.

Table 2: NARS LLR scoring parameters.

Parameter	Value	Description
Smoothing	Laplace $(+1)$	Pseudocount for LLR estimation
LLR clamp	$[-5, 5]$	Log-likelihood ratio bounds
Threshold τ	Calibrated	Median-midpoint (ICE-ID) or validation F_1 (classic ER)

These settings define the evidence weighting and decision boundary used by the matcher; the per-dataset threshold remains data-driven via calibration.

Judgment Vocabulary. The principal atomic judgments generated by the preprocessor:

- **Name**: `same_nafn_norm`, `different_nafn_norm`, `same_first_name`, `different_first_name`, `same_patronym`, `different_patronym`, `same_surname`, `different_surname`
- **Birthyear**: `same_birthyear`, `birthyear_very_close` ($\leq$2 yr), `birthyear_close` ($\leq$5 yr), `different_birthyear`
- **Demographics**: `same_sex`, `different_sex`, `same_marriagestatus`, `different_marriagestatus`
- **Geography**: `same_farm`, `same_parish`, `same_district`, `same_county`, `different_farm`, `different_parish`, `different_district`, `different_county`
- **Temporal**: `differ_in_X_years` (where X = |heimild$_1$ - heimild$_2$|)

The full list of emitted judgment tokens and the dataset-specific preprocessors are in the released code repository.

Complexity. In the LLR variant, matching scales as $O(N \cdot |J|)$ where N is the number of scored pairs and $|J|$ is the average number of generated judgments per pair.

3 Results

Evaluation protocols. We report results under two distinct evaluation regimes, which are *not directly comparable* in absolute terms:

1. **Pairwise benchmark** (Tables 3, 5, 6, 8): labeled pair classification on balanced samples (2:1 negative-to-positive ratio, capped at 50k pairs). ICE-ID uses fixed temporal train/validation/test splits by census year; classic ER datasets use their benchmark-provided splits. The ICE-ID positive rate is approximately 33.3% under the 2:1 sampling design.
2. **Candidate-graph evaluation** (Table 4): NARS scored on a candidate graph produced by token blocking, with realistic positive rate ($\approx$1%). Metrics include ranking (P@k, R@k) and clustering (ARI, B^3).

Table 3: ICE-ID pairwise results (F_1) from the benchmark runs, ordered by F_1. Ditto and HierGAT use their own data preparation and train/test splits. MatchGPT was evaluated on only 50 pairs.

Model	F_1	Note
Ditto	**0.997**	Separate eval
NARS	0.994	—
Fellegi–Sunter	0.994	—
Rules	0.990	—
AnyMatch	0.907	—
HierGAT	0.857	Separate eval
MatchGPT	0.276	50 pairs only

3.1 ICE-ID Results

Table 3 summarizes pairwise performance on ICE-ID. Under our shared evaluation protocol, NARS and Fellegi–Sunter both reach F_1=0.994; classical probabilistic linkage is surprisingly competitive on this well-structured dataset [6, 27, 21]. Ditto reports the highest F_1 (0.997) under its own data preparation and splits, while HierGAT (0.857) underperforms under its separate pipeline. AnyMatch transfers well without dataset-specific training (0.907). The NARS result (F_1=0.994, AUC=0.998) shows that symbolic judgment-based evidence captures most of the discriminative signal in the pairwise benchmark setting while preserving a fully decomposable score for each prediction. MatchGPT was evaluated on only 50 pairs per dataset, so its scores should be interpreted with caution.

3.2 Pairwise–Cluster Mismatch

Clustering and ranking metrics can diverge from pairwise classification metrics because they depend on the evaluation graph, not only on local pair decisions. In the pairwise benchmark, pairs are sampled independently rather than scored on a full deployment-style candidate graph, so connected-component clustering is computed on an induced sampled graph and cannot fully enforce transitivity. As a result, ARI-CC may differ substantially from what pairwise F_1 suggests. On ICE-ID, the pairwise benchmark still yields ARI-CC=0.881 (Table 6), but this should not be compared directly to the candidate-graph clustering results in Table 4. Likewise, P@k depends on score ordering, not just the decision threshold, so a model can achieve high F_1 while ranking pairs poorly.

A random scorer on the same sampled graph confirms this: it yields ARI-CC ≈ 0 and P@k $\approx$ the positive rate, matching the behavior expected under sparse sampling. This is a protocol artifact, not an implementation error.

Implication. High pairwise metrics do not guarantee good clustering or ranking performance. Entity resolution pipelines should be evaluated end-to-end on the downstream task (clustering on a candidate graph) rather than on pairwise classification alone. We present such a graph-based evaluation for NARS in the next section.

3.3 End-to-End Graph Evaluation (Ranking, Clustering, Cost)

To avoid overly optimistic pair sampling, we evaluate NARS on a candidate graph produced by token blocking [22, 3, 15], then compute ranking and clustering metrics on the resulting scored edges.

Table 4: NARS end-to-end evaluation on an ICE-ID candidate graph (token blocking, 407,934 scored pairs, τ=0.16, re-calibrated on candidate-graph validation scores). Ranking metrics use k equal to the number of positives. Clustering uses connected components on thresholded edges.

Pairs	Pos rate	F_1	AUC	P@k	R@k	ARI-CC	B^3 P	B^3 R	B^3 F_1
407,934	0.013	0.473	1.000	0.961	0.961	<0.001	0.218	0.997	0.358

Table 4 reports NARS on a realistic candidate graph (positive rate $\approx 1.3\%$) with a re-calibrated threshold (τ=0.16). Threshold-independent ranking metrics remain strong: P@k and R@k both reach 0.961, and AUC rounds to 1.000, which indicates excellent score ordering under heavy class

imbalance. In contrast, threshold-dependent pairwise F_1 drops to 0.473 because the low threshold required for high recall introduces many false positives (precision=0.310). B^3 metrics show the same trade-off: B^3 recall is 0.997, but B^3 precision is 0.218, indicating over-merged clusters. ARI-CC remains near zero because connected-component clustering on the thresholded graph is dominated by a few large merged components. These results show that NARS produces highly informative scores, but converting those scores into stable clusters on a realistic candidate graph requires graph-aware post-processing beyond simple thresholding.

3.4 Baseline Comparison

We evaluate NARS against several baselines on ICE-ID and standard ER datasets, following the evaluation methodology of Köpcke et al. [11] for real-world match problems. Table 5 compares NARS with deep learning matchers (Ditto [13], HierGAT [33]; see also 20 for the design space), unsupervised (ZeroER [32]), zero-shot (AnyMatch [34]), LLM-based (MatchGPT [25]; see also 12, 29), and supervised contrastive methods [24].

Table 5: Baseline comparison (F_1 scores) across ICE-ID and standard ER datasets. Each model uses its own evaluation pipeline (see provenance graph, Fig. 8); cross-model comparisons should be made with caution. MatchGPT was evaluated on 50 pairs per dataset. ZeroER is an unsupervised two-table matcher and is not applicable to ICE-ID's single-table census linkage task. Bold marks the best score among the evaluated runs shown in this table.

Model	ICE-ID	Abt–Buy	Amazon–Google	DBLP–ACM	DBLP–Scholar
NARS	0.994	**0.648**	**0.759**	**0.997**	**0.995**
Ditto	**0.997**	0.525	0.732	0.970	0.940
Fellegi–Sunter	0.994	0.126	0.351	0.961	0.776
HierGAT	0.857	0.000	0.577	0.940	0.973
ZeroER	—	0.420	0.393	0.991	0.817
AnyMatch	0.907	0.400	0.625	0.947	0.970
MatchGPT†	0.276	0.333	0.000	0.727	0.400

† Evaluated on 50 pairs per dataset; high variance expected.

Within the collected benchmark runs, NARS attains the highest reported F_1 on four of five datasets (Abt–Buy, Amazon–Google, DBLP–ACM, and DBLP–Scholar), although these comparisons remain sensitive to differences in preprocessing and evaluation pipelines. Ditto is the strongest neural matcher overall, achieving the best ICE-ID score (0.997) and competitive citation-matching results. HierGAT performs well on citation datasets (DBLP–ACM 0.940, DBLP–Scholar 0.973) and reaches 0.857 on ICE-ID, but collapses on Abt–Buy (F_1=0.000), likely because the training split contains only 49 positive pairs and the product fields are long, noisy, and often incomplete. ZeroER is competitive on citation matching without labels but is not applicable to ICE-ID's single-table setting. AnyMatch shows consistent zero-shot transfer across datasets. Fellegi–Sunter matches NARS on ICE-ID (0.994) but degrades substantially on two-table benchmarks where field semantics differ from its comparison rules. Figure 3 summarizes these trends.

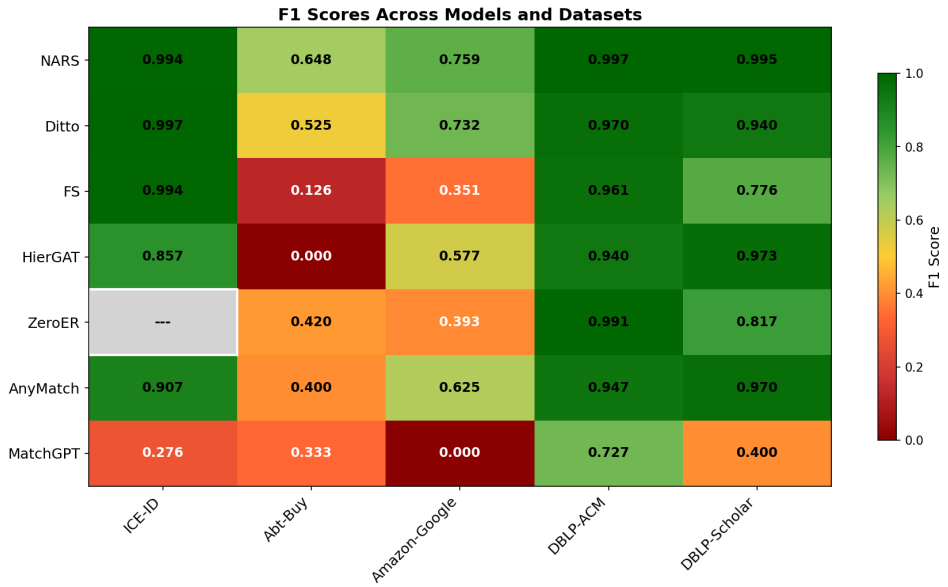

Figure 3: F_1 heatmap across models and datasets from the benchmark runs.

Figure 4 summarizes NARS performance across ICE-ID and the classic ER suite.

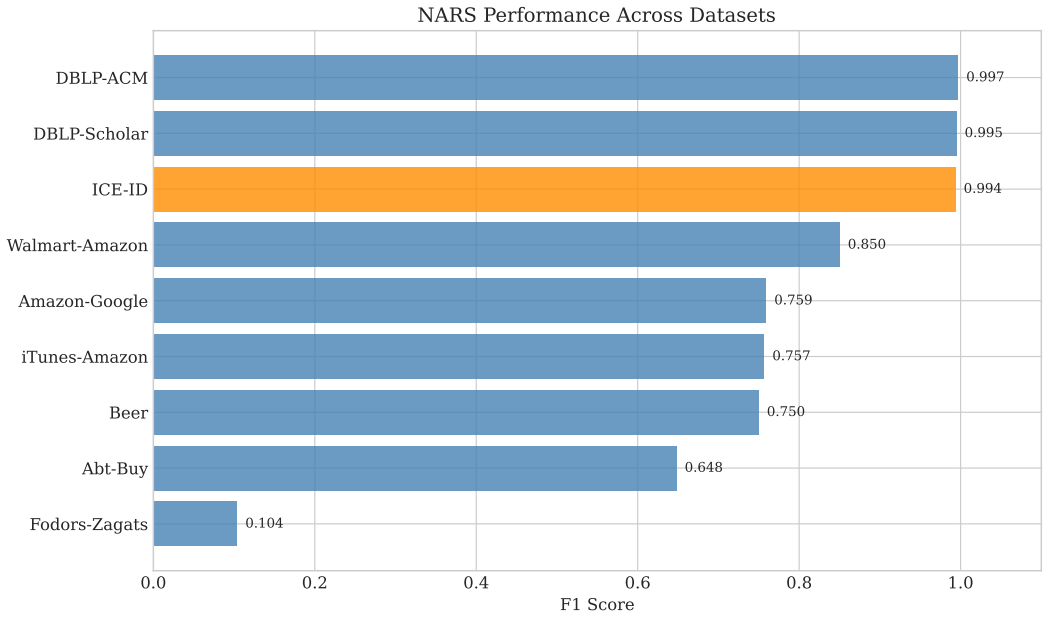

Figure 4: NARS F_1 across ICE-ID and classic ER datasets. ICE-ID achieves F_1=0.994 under strict temporal splitting with hard negatives; classic two-table performance remains sensitive to dataset size and label density, with small-test datasets showing higher variance.

Three performance tiers are visible. The citation datasets DBLP–ACM (0.997) and DBLP–Scholar (0.995) sit at the top alongside ICE-ID (0.994): all three share well-structured fields with high discriminative power (title, author, year), allowing the LLR judgments to separate matches from non-matches with near-perfect accuracy. A middle tier spanning Walmart–Amazon (0.850), Amazon–Google (0.759), iTunes–Amazon (0.757), and Beer (0.750) reflects the greater difficulty of product and media matching, where descriptions are noisier, field alignment is less consistent, and positive-pair rates are low (0.4–7%). Abt–Buy (0.648) falls below this group, likely because its unstructured

product text reduces the effectiveness of field-level comparisons. The outlier is Fodors–Zagats (0.104), where only four positive test pairs produce an extremely low-precision result: the model identifies all true matches (recall = 1.0) but generates many false positives, as the small label set provides insufficient signal for threshold calibration.

3.5 Performance on Other Datasets

We also evaluated the performance of the NARS method against SOTA across standard two-table ER datasets (Abt–Buy, Amazon–Google, DBLP–ACM, DBLP–Scholar, iTunes–Amazon, Walmart–Amazon, Beer, Fodors–Zagats).

Table 6: NARS results across all datasets (see Tab. 10 for metric definitions). Beer and iTunes–Amazon have <20 test positives (9 and 16), yielding higher-variance metrics.

Dataset	P	R	F_1	Acc	Thr	AUC	ARI-CC	ARI-AG	P@k	R@k
ICE-ID	**0.995**	**0.992**	**0.994**	**0.996**	0.50	**0.998**	**0.881**	**0.413**	**0.995**	**0.995**
Abt–Buy	0.522	0.854	0.648	0.938	0.28	0.945	0.528	0.739	0.616	0.616
Amazon–Google	0.720	0.802	0.759	0.962	0.29	0.967	0.786	0.815	0.743	0.743
DBLP–ACM	1.000	0.994	0.997	0.999	0.95	1.000	1.000	1.000	1.000	1.000
DBLP–Scholar	0.998	0.993	0.995	0.996	0.23	1.000	0.998	0.984	0.993	0.993
iTunes–Amazon	0.667	0.875	0.757	0.996	0.95	0.999	0.807	0.737	0.875	0.875
Walmart–Amazon	0.817	0.887	0.850	0.980	0.94	0.996	0.879	0.897	0.879	0.879
Beer	0.857	0.667	0.750	0.998	0.92	0.973	0.917	0.917	0.667	0.667
Fodors–Zagats	0.055	1.000	0.104	0.954	0.95	0.986	0.077	0.137	0.000	0.000

Table 7: Prior supervised SOTA F_1 vs NARS on standard ER benchmarks. ΔF_1 = NARS F_1 – SOTA F_1. ICE-ID is omitted as a new benchmark with no prior published result. SOTA sources: [a] R-SupCon [24], [b] Ditto [13].

Dataset	SOTA F_1	Source	NARS F_1	ΔF_1
DBLP–ACM	0.990[b]	Ditto	0.997	+0.007
DBLP–Scholar	0.956[b]	Ditto	0.995	+0.039
Walmart–Amazon	0.868[b]	Ditto	0.850	−0.018
Amazon–Google	0.793[a]	R-SupCon	0.759	−0.034
iTunes–Amazon	0.971[b]	Ditto	0.757	−0.214
Beer	0.944[b]	Ditto	0.750	−0.194
Abt–Buy	0.943[a]	R-SupCon	0.648	−0.295
Fodors–Zagats	1.000[b]	Ditto	0.104	−0.896

With the benchmark-specific preprocessing described in the appendix, NARS matches or exceeds reported supervised SOTA on DBLP–ACM (+0.007) and DBLP–Scholar (+0.039), and remains competitive on Walmart–Amazon (−0.018). On product-matching datasets with long unstructured text (Abt–Buy, Amazon–Google) and small-sample datasets (Beer, iTunes–Amazon) the gap widens, reflecting the limits of field-level symbolic judgments on noisy text. Fodors–Zagats is an outlier: with only four positive test pairs, the F_1 of 0.104 is driven by low precision despite perfect recall, as discussed in the Figure 4 analysis above.

Table 6 reports NARS pairwise performance on ICE-ID and standard two-table ER datasets. For two-table benchmarks, ARI values are computed on induced clusters from thresholded pair graphs; they are included for comparability but should be interpreted more cautiously than pairwise F_1/AUC. The deployment-faithful graph view for ICE-ID is reported separately in Table 4.

3.6 Ablation Study: Judgment Type Contributions

To understand which judgment types contribute most to NARS's performance, we conducted ablation experiments removing each category of judgments from the LLR-based scoring pipeline. These ablations use the same evaluation protocol as the primary benchmark (Table 6): identical pair sampling, train/validation/test splits, and threshold calibration, so absolute F_1 values are directly comparable.

Table 8: NARS ablation on ICE-ID (same protocol as primary benchmark). Each row removes one judgment category. ΔF_1 is relative to the full model.

Ablation	$\mathbf{F_1}$	$\Delta\mathbf{F_1}$	**AUC**
Full model	0.994	—	0.998
− Name judgments	0.916	−0.078	0.996
− Geographic judgments	0.990	−0.004	0.998
− Birthyear judgments	0.994	<0.001	0.998
− Sex judgments	0.994	0.000	0.999
− Census year (heimild)	0.994	0.000	0.998

Name judgments are the most informative feature category ($\Delta F_1 = -0.078$), primarily by reducing recall from 0.992 to 0.859 when removed. Geographic judgments have a small but measurable effect ($\Delta F_1 = -0.004$). The remaining categories—birthyear, sex, and census year—show negligible individual impact, indicating substantial redundancy among features. The high baseline performance (F_1=0.994) means the model is already near ceiling, which compresses ablation deltas; the relative ordering of feature importance is more informative than the absolute magnitudes. Figure 5 visualizes these results.

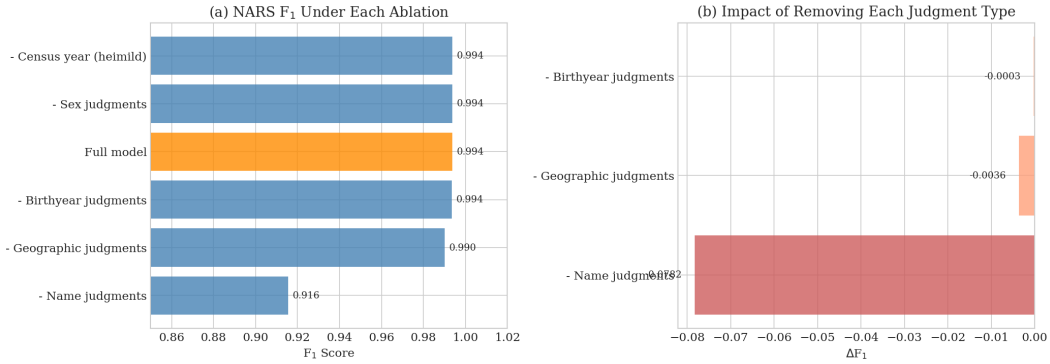

Figure 5: (a) NARS F_1 under each ablation. (b) Impact (ΔF_1) of removing each judgment type, ordered by magnitude.

3.7 Interpreting NARS Decisions

A key advantage of the LLR scoring framework is that every match decision decomposes additively into per-judgment contributions, each with a clear semantic meaning. Table 9 lists the learned LLR weights for the ICE-ID judgment vocabulary, sorted by magnitude.

Table 9: Learned LLR weights on ICE-ID (top 14 by $|w|$), estimated from the ICE-ID training split. $P(j \mid M)$ and $P(j \mid \neg M)$ are empirical judgment rates in matching and non-matching training pairs. Weights are clamped to $[-5, +5]$.

Judgment	**LLR** w	$P(j\|M)$	$P(j\|\neg M)$
`same_nafn_norm`	$+5.00$	0.971	0.001
`same_farm`	$+5.00$	0.419	<0.001
`different_birthyear`	-5.00	0.003	0.909
`different_first_name`	-5.00	0.007	0.987
`different_sex`	-5.00	0.002	0.501
`same_parish`	$+4.82$	0.644	0.005
`same_county`	$+4.51$	0.682	0.008
`different_patronym`	-4.37	0.012	0.962
`same_first_name`	$+4.30$	0.994	0.014
`same_birthyear`	$+4.10$	0.578	0.010
`same_patronym`	$+3.73$	0.983	0.023
`different_nafn_norm`	-3.54	0.029	0.999
`same_district`	$+2.71$	0.800	0.053
`birthyear_very_close`	$+2.52$	0.412	0.033

The weight magnitudes are consistent with the ablation results (Table 8): name and patronym judgments dominate, followed by geographic and birthyear evidence. Five judgments hit the ± 5 clamp, indicating that the unclamped LLR is even more extreme—for instance, `same_farm` appears in 41.9% of matches but less than 0.1% of non-matches. The geographic judgments also receive strong positive weights, but their exact ordering is dataset-specific and reflects how discriminative each level is under the ICE-ID sampling and preprocessing setup. The remaining judgments have smaller magnitudes and cover census proximity, sex, and disagreement cases; the full set is released with the code.

Per-pair decomposition. Figure 6 illustrates the additive score decomposition for a true-positive pair from the ICE-ID test set. The pair consists of two census records for *Áslaug Bjarnadóttir*: one from the 1855 census (birth year 1843) and one from the 1860 census (birth year 1844), both at farm 664 in the same parish. Each bar shows one judgment's contribution to the log-odds under the learned LLR weights. The sum of all contributions ($+34.07$) plus the base-rate prior (-0.69) yields total log-odds of $+33.38$; applying the sigmoid gives a match probability of $\sigma(+33.38) \approx 1.000$.

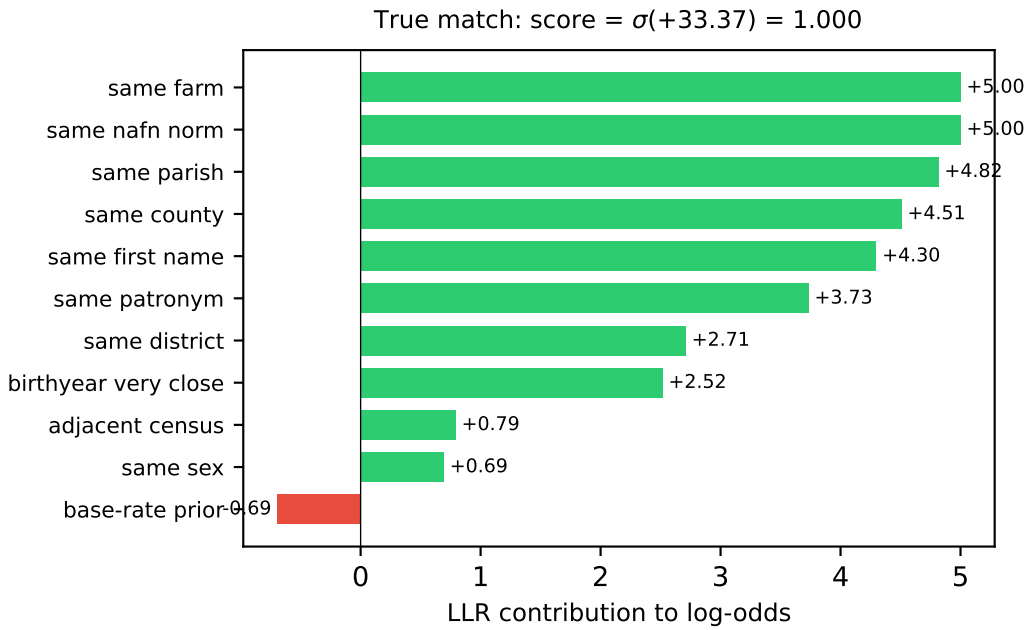

Figure 6: Per-judgment LLR decomposition for a true-positive ICE-ID pair (Áslaug Bjarnadóttir, census 1855 vs. 1860). Green bars contribute evidence *for* a match; red bars contribute evidence *against*. The total log-odds ($+33.38$) produces $\sigma(+33.38) \approx 1.000$.

Error diagnosis via decomposition. Figure 7 shows the same decomposition for a *false positive*: two different men both named *Magnús Magnússon* in the 1835 census, living in different counties with a 20-year birth-year gap (1809 vs. 1829). Three name judgments (`same_nafn_norm` $+5.00$, `same_first_name` $+4.30$, `same_patronym` $+3.73$) contribute $+13.03$ to the log-odds, overwhelming the geographic and birth-year counter-evidence (-10.92). The net log-odds of $+2.12$ yields $\sigma(+2.12) = 0.893$, above the threshold. The decomposition immediately diagnoses the failure mode: Icelandic patronymic naming produces common name collisions ("Magnús, son of Magnús"), and the current judgment vocabulary does not discount high-frequency names. A domain expert inspecting this breakdown could propose frequency-weighted name judgments or a stronger birth-year penalty as targeted fixes—a debugging workflow that is not available when the scorer is a black-box neural model.

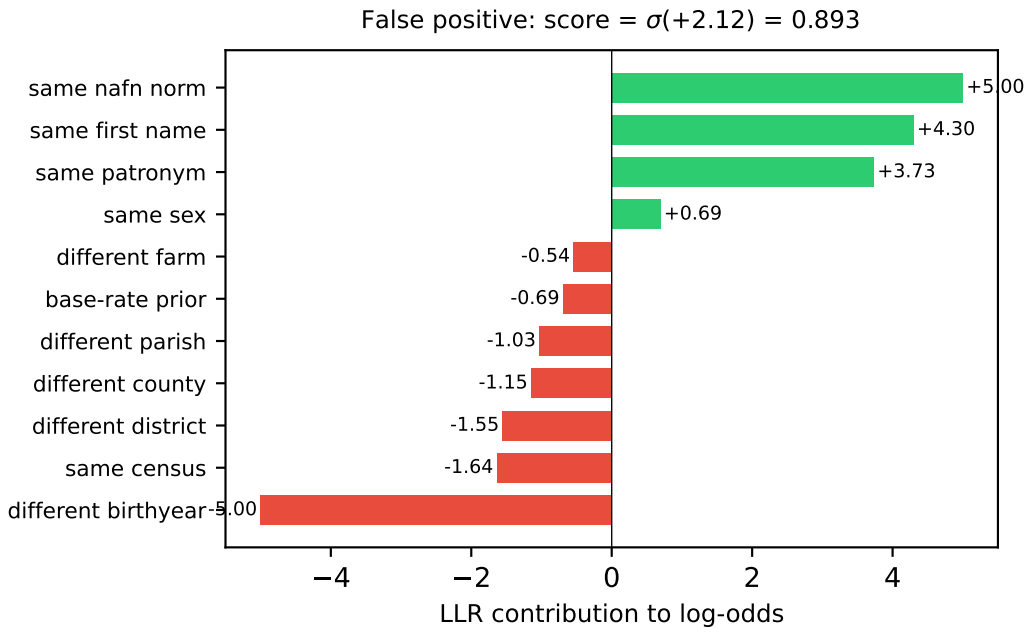

Figure 7: Per-judgment LLR decomposition for a false-positive ICE-ID pair (two distinct individuals both named Magnús Magnússon, census 1835). Name evidence overwhelms geographic and birth-year counter-evidence, producing $\sigma(+2.12) = 0.893$.

This decomposition contrasts with neural matchers, where the match probability is produced by a learned latent scoring function rather than an explicit symbolic evidence sum. Ditto outputs a single probability from a fine-tuned language model, and HierGAT produces scores through learned graph-based representations. In such models, pair-level explanations are typically obtained through additional interpretation methods (for example, attention inspection or feature-attribution tools) rather than directly from the deployed scoring function. NARS provides this decomposition natively: the LLR weight of each judgment is the exact contribution that judgment makes to the final log-odds, and any decision can be audited by inspecting the breakdown. This property is particularly valuable in historical record linkage, where scholars need to understand and validate linkage decisions before drawing substantive conclusions about demographic or genealogical patterns.

4 Discussion & Conclusion

Our calibrated NARS pipeline (median-midpoint thresholding on ICE-ID and validation-F_1 thresholding on classic ER datasets) shows that a symbolic, evidence-based matcher can achieve competitive pairwise accuracy in entity resolution without deep representation learning or GPU resources, while keeping the deployed scoring path directly inspectable. On the balanced ICE-ID pairwise benchmark, NARS reaches F_1=0.994, matching Fellegi–Sunter and trailing Ditto (0.997) by a small margin under separate evaluation pipelines. Across the collected benchmark runs, NARS records the highest F_1 on four of five reported cross-dataset evaluations, and on the standard ER benchmarks it matches or exceeds prior reported supervised results on DBLP–ACM and DBLP–Scholar. The main advantage is not uniform dominance in F_1, but that each decision is expressed as an auditable sum of named evidence terms with explicit weights.

Fellegi–Sunter matches NARS on ICE-ID (0.994) but degrades on two-table benchmarks (e.g., 0.126 on Abt–Buy vs. NARS's 0.648). This contrast highlights the role of domain-adaptive preprocessing in NARS: dataset-specific symbolic judgments transfer across ER schemas more effectively than a fixed set of classical comparison rules designed for a structured census setting.

Our results also reveal important limitations. The candidate-graph evaluation (Table 4) shows that NARS produces excellent score rankings (AUC rounds to 1.000; P@k=R@k=0.961), but converting those scores into hard decisions on a realistic graph (positive rate ≈1.3%) remains difficult: threshold-

dependent F_1 drops to 0.473 and connected-component clustering over-merges (B^3 precision=0.218). This gap between ranking quality and hard-decision quality is a known entity-resolution challenge [3] and motivates graph-aware post-processing, including Bayesian partition priors [14], knowledge-driven graph models [10], and progressive pay-as-you-go strategies [31]. Learned blocking schemes [15] may also improve candidate generation.

The ablation results (Table 8) show that name judgments are by far the most informative feature category ($\Delta F_1 = -0.078$), while geographic judgments have a small effect and the remaining categories (birthyear, sex, census year) show negligible individual impact at the high baseline F_1=0.994. The substantial redundancy among non-name features suggests that the model relies primarily on name matching, with other features serving as tiebreakers.

In conclusion, NARS demonstrates that symbolic reasoning can serve as a transparent, competitive baseline for entity resolution. As shown in Section 3.7, the deployed LLR scorer supports exact per-judgment decomposition of each prediction, which lets domain experts inspect and audit individual match decisions directly. Combined with low resource requirements, this makes NARS a useful complement to neural matchers in hybrid ER pipelines, especially in settings where auditability and error analysis are part of the core workflow.

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A Appendix / supplemental material

A.1 Artifact Provenance Graph

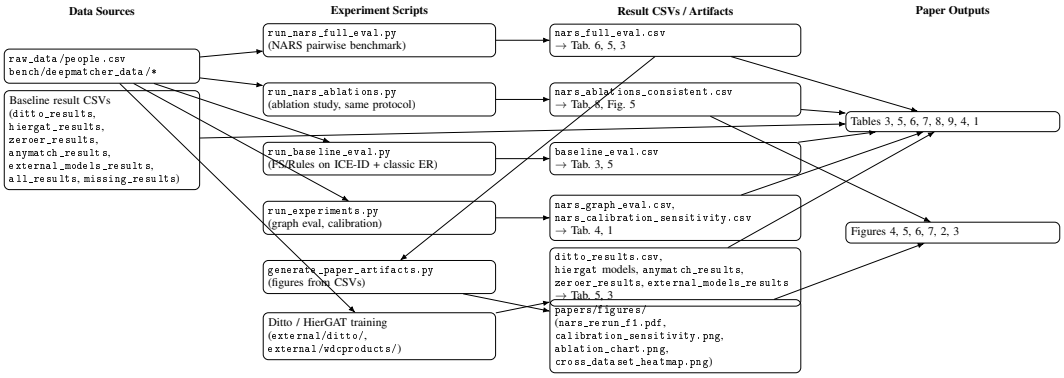

Figure 8: Provenance graph mapping data sources, scripts, result artifacts, and paper tables/figures. NARS, Fellegi–Sunter, and Rules results are produced by dedicated evaluation scripts; Ditto and HierGAT ICE-ID results come from direct training; other baseline results are from pre-computed CSVs.

A.2 Metrics

Metric	Definition
Precision (P)	$\frac{\text{TP}}{\text{TP}+\text{FP}}$ — fraction of predicted matches that are correct.
Recall (R)	$\frac{\text{TP}}{\text{TP}+\text{FN}}$ — fraction of true matches that are found.
F_1 Score	$2 \cdot \frac{P \cdot R}{P+R}$ — harmonic mean of precision and recall.
Accuracy (Acc)	$\frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}}$ — proportion of correct predictions.
Threshold (Thr)	Calibrated score cutoff τ separating matches vs. non-matches.
AUC	Area under the ROC curve — ranking quality over all thresholds.
ARI-CC	Adjusted Rand Index on connected-component clustering of thresholded graph.
ARI-AG	Adjusted Rand Index on agglomerative clustering of pairwise scores.
Precision@k (P@k)	Precision among the top-k highest-scoring pairs, where k equals the number of positives in slice.
Recall@k (R@k)	Recall among the top-k highest-scoring pairs.

Table 10: Definitions of the evaluation metrics used in Tables 6 and 7.

A.3 Benchmark-specific NARS preprocessing (classic ER datasets)

ICE-ID preprocessing remains unchanged and follows the rules described earlier in the paper. For the classic two-table ER benchmarks, we replaced the generic token-overlap preprocessing with dataset-specific judgments so that NARS can use the fields that define matches in each dataset.

Shared normalization and scoring. Text fields are lowercased, stripped, and normalized by removing non-alphanumeric characters and collapsing whitespace. Similarity is the maximum of token Jaccard overlap and SequenceMatcher ratio. For each text field we emit: `field_exact` if normalized strings match, else `field_sim_high` (≥ 0.90), `field_sim_med` (≥ 0.75), `field_sim_low` (≥ 0.50), or `field_sim_vlow` (otherwise). Numeric fields are parsed by extracting the first numeric token. We emit `field_exact` if absolute difference ≤ 0.01, `field_close` if absolute difference $\leq$ a dataset-specific threshold or relative difference ≤ 0.05, and `field_far` if relative difference ≥ 0.20. Year fields are parsed from 4-digit years and emit `field_same` (diff=0), `field_close` (diff ≤ 1), or `field_far`. Time fields accept seconds or `mm:ss` and emit `time_same` (diff=0), `time_close` (diff $\leq$ 5s), or `time_far`. Phone fields compare digits only and emit `phone_exact`, `phone_last7`, or `phone_mismatch`. Address numbers emit `addr_num_match` or `addr_num_diff` when parseable.

Dataset-specific judgments.

- **Abt–Buy:** `name`, `description` (text), `price` (numeric; abs ≤ 1.0 or rel ≤ 0.05).
- **Amazon–Google:** `title`, `manufacturer` (text), `price` (numeric; abs ≤ 1.0 or rel ≤ 0.05).
- **DBLP–ACM / DBLP–Scholar:** `title`, `authors`, `venue` (text), `year` (same/close/far).
- **iTunes–Amazon:** `Song_Name`, `Artist_Name`, `Album_Name`, `Genre` (text), `Price` (numeric; abs ≤ 0.5 or rel ≤ 0.05), `Time` (time), `Released` and `CopyRight` (year).
- **Walmart–Amazon:** `title`, `category`, `brand`, `modelno` (text), `price` (numeric; abs $\leq$ 1.0 or rel ≤ 0.05).
- **Beer:** `Beer_Name`, `Brew_Factory_Name`, `Style` (text), `ABV` (numeric; abs ≤ 0.3 or rel $\leq$ 0.05).
- **Fodors–Zagats:** `name`, `addr`, `city`, `type`, `class` (text), plus phone and address-number judgments.

Data alignment for DeepMatcher splits. For these benchmarks, pair IDs correspond to the source tables (e.g., `abt.csv`/`buy.csv`, `amazon.csv`/`google.csv`), so we align record IDs to the pair indices before applying the standard right-table offset to make IDs globally unique. This ensures that all training and evaluation pairs map to actual records.

Thresholding. ICE-ID uses the original median-based separation of positive/negative validation scores. For classic ER benchmarks, we select a per-dataset threshold that maximizes validation F_1 to accommodate extreme class imbalance.

A.4 NARS Background

NARS [30] uses *channels* to transform external information into Narsese (its internal knowledge representation), as implemented in OpenNARS [8]. A channel can process information of any modality—strings for natural language, matrices for visual signals—under a single principle: compounding atomic inputs (e.g., words in a sentence or pixels in an image) into structured terms. For the problem discussed here, channels process census records.

Compounding is guided by NARS's memory-based attention allocation rather than brute-force enumeration. However, when only census data are provided, NARS lacks the semantics of domain concepts (e.g., farm locations), so additional preprocessing of raw records is required. Future work could supply spatial and temporal relationships directly, allowing NARS to ground input semantics through geographical and historical context.

In summary, our approach adapts the compound generation method from NARS channels and the truth-value calculus to entity resolution, while omitting memory management and goal-driven inference components. This focused adaptation enables direct comparison with standard ML baselines on ER benchmarks.

A.4.1 Channel

The NARS channel maintains a short-term cache of recent inputs, each consisting of multiple atomic sensations. From this cache the channel constructs two types of compounds: *spatial compounds*, which group co-occurring sensations into a single moment, and *temporal compounds*, which capture implication relationships between spatial compounds for prediction. Because census records lack short-term sequential structure, we use only spatial compounds. Rather than enumerating all combinations of atomic sensations, the channel forms a compound as a whole and then determines which parts can be eliminated.

A.4.2 Truth-Value

In NARS, the truth-value of a judgment quantifies both positive and negative evidence [30]. Given w_+ units of positive evidence and w_- units of negative evidence, the truth-value (f, c) is defined as $f = w_+/(w_+ + w_-)$ and $c = (w_+ + w_-)/(w_+ + w_- + k)$, where $f \in [0, 1]$ (the *frequency*) is the proportion of positive evidence and $c \in [0, 1]$ (the *confidence*) measures how much existing evidence there is relative to k hypothetical future observations. Higher f indicates stronger positive support; higher c indicates that f is less likely to change with new evidence.

A.4.3 The Expectation of Truth-Values

Because the truth-value is two-dimensional, NARS defines a scalar summary called *expectation*: $e = c(f - 0.5) + 0.5$, with $e \in [0, 1]$. Values of e near 1 indicate a judgment that is both true and reliable; values near 0 indicate a judgment that is both false and reliable. When $e \approx 0.5$, the system is either ignorant ($f \approx 0.5$) or lacks sufficient evidence (c is low).

A.4.4 The Revision of Truth-Values

When two independent sources provide truth-values for the same judgment, NARS integrates them via *revision*: $w_+ = w_+^{(1)} + w_+^{(2)}$, $w_- = w_-^{(1)} + w_-^{(2)}$, with f and c recomputed from the pooled evidence counts.

A.4.5 Pilot Hyperparameter Tuning

Note: this subsection describes an early pilot procedure used to select NARS hyperparameters before the final benchmark protocol (Section 2) was fixed.

Census records up to 1899 served as training data and records after 1899 as test data. Records were sorted by individual so that records from the same person appeared consecutively. For each record,

m subsequent rows were paired with it, yielding both positive pairs (same person) and negative pairs (different persons).

At test time, x individuals were sampled from the test set and all pairwise match scores among their records were computed. A symmetric threshold δ around 0.5 was swept: pairs with score $> 0.5 + \delta$ were predicted as matches and pairs with score $< 0.5 - \delta$ as non-matches, and F_1 was computed for each δ. Default settings: $1/10{,}000$ of training data, $x = 1{,}000$ test individuals, $m = 5$ neighbor rows, and 10 reference patterns.

Increasing the number of reference patterns improved reliability, while varying the training proportion and observation scope had less consistent effects.

Handcrafted Rules (pilot-era rule set). The following field-level judgments were used in early pilot experiments. The final benchmark uses the extended judgment vocabulary described in Section 2 (which adds graded birthyear thresholds, geographic hierarchy levels, and temporal gap judgments). When processing a record pair, the pilot rules generate:

Heimild: Emits `differ_in_X_years`, where X is the absolute difference in census year between the two records.

nafn_norm: Emits `same_nafn_norm` or `different_nafn_norm` based on whether the normalized full names match. The same logic applies to ***first_name***, ***patronym***, and ***surname***.

Birthyear: Emits `same_birthyear` or `different_birthyear` based on whether the birth years match.

Sex: Emits `same_sex` or `different_sex` based on whether the sex values match.

Status: Because social status is categorical rather than binary, two judgments are emitted: `status_is_X` and `status_is_Y`, where X and Y are the status values from each record (unordered).

Marriagestatus: Emits `same_marriagestatus` or `different_marriagestatus` based on whether the marital status values match.

Farm: Emits `same_farm` or `different_farm` based on whether the farm IDs match. The same logic applies to ***county***, ***parish***, and ***district***.

Label: The ground-truth indicator: records are from the same individual if and only if they share the same `person` ID.